Function Context: A2

Function Background

This function comes from the input-validation module of scikit-learn. Machine-learning algorithms often require input data to satisfy specific constraints on data type, dimensionality, and values. The function checks an input array and normalizes it when necessary.

Input and Output

Input: array, the data to be checked; the remaining parameters specify permitted input forms and constraints.

Output: The array after validation and any necessary conversions.

Exceptions: The function raises an exception when the input does not meet the requirements.

Key Terms

Term	Meaning
array-like	A data object that can be converted to an array.
sparse matrix	A matrix that stores only non-zero elements.
dtype	The data type of array elements.
ndim	The number of dimensions of an array.
finite value	A finite numeric value, excluding invalid values such as positive or negative infinity.
pandas Series	A one-dimensional data structure in pandas.
writeable	Whether the underlying array data can be modified.

Helper Variables and Functions

Name	Brief Description
sp.issparse(array)	Checks whether the input is a sparse matrix.
_ensure_sparse_format(...)	Checks and normalizes the format of a sparse matrix.
_assert_all_finite(...)	Checks whether an array contains disallowed non-finite values.
_num_samples(array)	Gets the number of samples in an array.
check_non_negative(...)	Checks whether an array contains negative values.
np.may_share_memory(...)	Checks whether two arrays may share underlying memory.

Background Not Required

Participants do not need to be familiar with the model-training process in scikit-learn. They only need to use the code to determine which checks and conversions are applied to different input forms.